E7 Introduction to Computer Programming for Scientists and Engineers

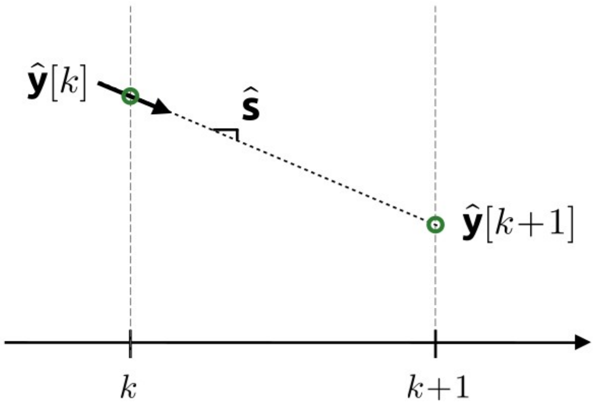
Lecture week 14-B

Numerical methods for ODEs (continued)

Berkeley ENGINEERING

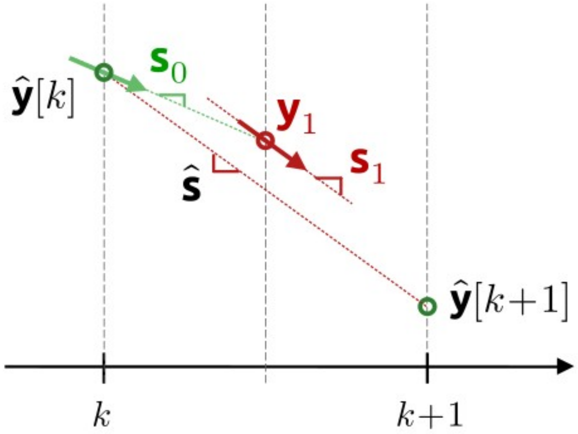
Euler's method

$$\hat{\mathbf{s}} = f(\hat{\mathbf{y}}[k], t)$$



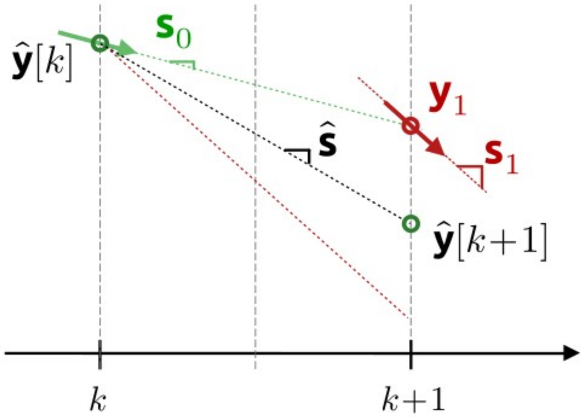
Midpoint method

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$
$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + (h/2)\mathbf{s}_0$$
$$\mathbf{s}_1 = f(\mathbf{y}_1, t + h/2)$$
$$\hat{\mathbf{s}} = \mathbf{s}_1$$



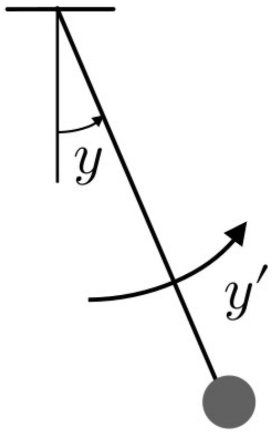
Heun's method

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$
$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + h\,\mathbf{s}_0$$
$$\mathbf{s}_1 = f(\mathbf{y}_1, t + h)$$
$$\hat{\mathbf{s}} = 0.5(\mathbf{s}_0 + \mathbf{s}_1)$$



Example: Simple pendulum

$$y'' + ay'\,|y'\,| + b\,\mathsf{sin}(y) = 0$$



Runge-Kutta 4

$$\mathbf{s}_0 = f(\hat{\mathbf{y}}[k], t)$$

$$\mathbf{y}_1 = \hat{\mathbf{y}}[k] + (h/2)\ \mathbf{s}_0$$

$$\mathbf{s}_1 = f\left(\mathbf{y}_1\ ,\ t + h/2\right)$$

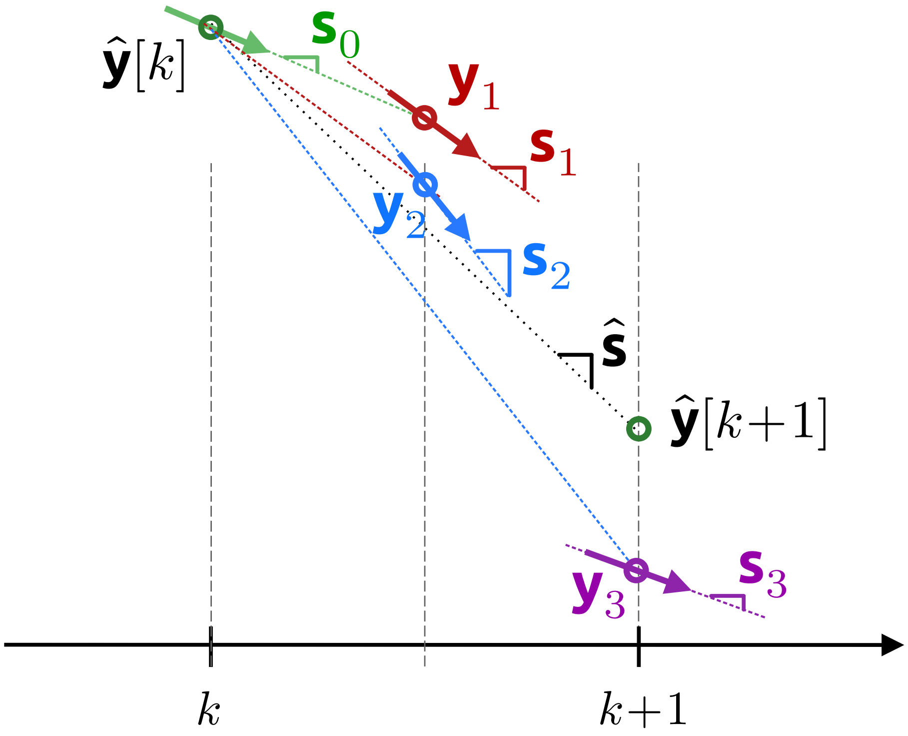
$$\mathbf{y}_2 = \hat{\mathbf{y}}[k] + (h/2)\ \mathbf{s}_1$$

$$\mathbf{s}_2 = f\left(\mathbf{y}_2\ ,\ t + h/2\right)$$

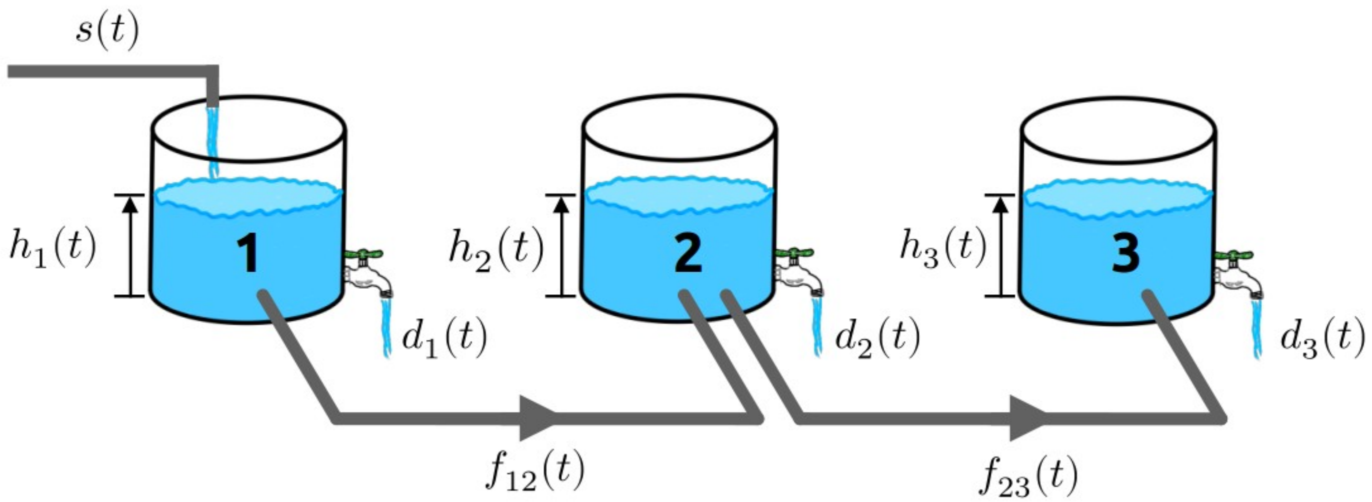
$$\mathbf{y}_3 = \hat{\mathbf{y}}[k] + h\ \mathbf{s}_2$$

$$\mathbf{s}_3 = f\left(\mathbf{y}_3\ ,\ t + h\right)$$

$$\hat{\mathbf{s}} = (1/6)\left(\mathbf{s}_0 + 2\ \mathbf{s}_1 + 2\ \mathbf{s}_2 + \mathbf{s}_3\right)$$



Example: Interconected water tanks



An automatic water balancing system sets the flows between the tanks to:

$$f_{12}(t) = \alpha_1(h_1(t) - h_2(t))$$
$$f_{23}(t) = \alpha_2(h_2(t) - h_3(t))$$